

Comprehensive Introduction Document for J2OS Software Development Infrastructure

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Audience: Organizations, Public and Private Companies, IT Managers, and Development Teams

Introduction

In the fast-paced and competitive world of information technology, organizations are seeking solutions to produce secure, scalable, and efficient software with minimal cost and time. The J2OS software development infrastructure, an advanced and integrated product from Java School, is designed to simplify the development process, reduce reliance on complex coding, and enhance productivity. Built on a Service-Oriented Architecture (SOA) and a Low-Code approach, this product provides automated and intelligent tools, enabling organizations to implement complex systems with high speed and security.

History and Development Goals

With over 20 years of Java School's experience in Java programming and cutting-edge technologies, J2OS was developed as a comprehensive solution to address the challenges of software production in large organizations. The primary goal of this product is to create a unified and standardized infrastructure that resolves common issues such as code inconsistencies, dependency on developer skills, and security and management complexities.

Key Features of J2OS

1. Reduced Coding and Accelerated Development

- At least 25% reduction in coding volume: Simplifies development and maintenance processes.
- At least 25% increase in project development speed: Automates repetitive and time-consuming tasks.
- Code Integration: Ensures harmony among system components.
- Low-Code Approach: Enables rapid development with minimal manual coding.

2. Dynamic Access Management

- Dynamic, code-free management of users, roles, and groups.
- Dynamic configuration of access to panels, fields, operations, and records.

3. **Advanced Built-in Security**

- Automatic prevention of XSS and SQL Injection attacks without coding.
- Data encryption to protect sensitive information.
- Security monitoring, access monitoring, and cyber threat management.

4. **Process Automation**

- Automatic logging of user and system activities (User & System Logging).
- Automatic implementation of pagination and table search capabilities.
- Automated concurrency management to prevent transaction conflicts.
- Automatic resolution of dependency and entity relationship issues.

5. **Independent and Distributed Engines**

- Independent Process Engine: Based on a distributed architecture, compatible with BPMN 2.0 (e.g., Camunda).
- Reporting Engine: Compatible with Jasper Report, offering three report types:
 - Designed with SQL in Jasper Studio.
 - Designed without SQL in Jasper Studio.
 - Automated reports using SQL.

6. **Modern and Efficient Architecture**

- Microservices Architecture (SOA): Provides flexibility and scalability.
- Cloud-Native Architecture Support: Simplifies deployment in cloud environments.
- High processing speed and easier debugging.
- Simple software distribution.

7. **AI and Data Mining Capabilities**

- Integration of artificial intelligence and data mining features for data analysis and improved decision-making.

Competitive Advantages of J2OS

- **For Organizations:**
 - Reduced development and maintenance costs.
 - Faster project delivery.
 - High security and reduced cybersecurity risks.

- **For Development Teams:**
 - Simplified coding and project management complexity.
 - Automated tools to streamline processes.
 - Automatic resolution of technical challenges like concurrency and dependencies.

- **For Managers:**
 - Comprehensive, real-time reporting.
 - Precise access control and process oversight.
 - Analytical tools for better organizational management.

J2OS Technical Infrastructure

1. **Core Components:**
 - Infrastructure Core: Manages entities, security, and transactions.
 - Process Engine: Supports organizational processes with a distributed architecture.
 - Report Generator Engine: Produces dynamic and static reports.

2. **Tools Utilized:**
 - Spring Framework: Base framework.
 - Hibernate ORM (JPA): Database framework.
 - JTA: Transaction management.
 - Camunda BPMN Engine: Process engine.
 - Jasper Report Libraries: Reporting.

- Cherry Framework: Complementary framework.
- Jakarta EE Security: Authentication.
- Databases: Oracle, PostgreSQL, MongoDB, and Elasticsearch.

3. Architecture and Deployment:

- Supports internal MVC architecture and external Enterprise Service-Oriented architecture.
- Service distribution capability with NGINX for enhanced performance.
- Compatibility with enterprise cloud environments.

J2OS Deployment Process

1. Analysis and Requirement Gathering:

- Identifying organizational needs and assessing existing infrastructure.

2. Setup and Initial Configuration:

- Infrastructure installation and organizational structure definition (users, roles, groups).

3. Training and Knowledge Transfer:

- Training for development teams and end-users.
- Provision of technical and educational documentation.

4. Support and Maintenance:

- 24/7 support and periodic updates.

Costs

Deployment and training costs depend on the organization's scale and custom requirements. Suggested cost examples:

- Infrastructure deployment: Based on organizational needs.
- Developer team training (20 people, 50 hours).

- End-user team training (20 people, 16 hours).
- Periodic support and updates.

Conclusion

The J2OS software development infrastructure introduces a revolutionary approach to building organizational systems by offering an integrated, secure, and rapid solution. With a 25% reduction in coding volume, a 25% increase in development speed, and automation of key processes, J2OS empowers organizations to achieve their goals with minimal complexity. J2OS is not just a software infrastructure—it's a reliable partner for your organization's technological future.

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